PROBLEM CL6

Elbow Discharging a Vertical Jet: Velocity, Force and Heat

Original statement · CL6.pdf

Elbow | Linear momentum | Energy | Heat through the walls

OFFICIAL PROBLEM STATEMENT

A liquid of density ρ and specific heat c flows through the elbow-shaped pipe shown in CL6.pdf, discharging to the atmosphere as a vertical jet through an outlet section of surface area A_o, different from the value A_i at the inlet. Gravity is assumed to have a negligible effect on the fluid motion. Besides, uniform flow properties are assumed at the inlet and outlet sections, where readings from a thermometer give different values T_i and T_o for the temperature. The velocity at the outlet section is measured to be v_o and a manometer is used to determine the value p_i for the pressure at the inlet section. In terms of all of these quantities, determine:

- The value of the velocity at the inlet v_i.
- The two components of the force (F_x, F_z) exerted on the pipe.
- The amount of heat Q transferred per unit time to the liquid by heat conduction through the pipe walls.

Source: CL6.pdf (page 1), also solved in class (class notes, CL6). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid, $e = cT$; neglect $\bar{g}$; uniform properties at inlet and outlet (no viscous stresses or conduction across them).
- Free jet: $p = p_a$ over the outlet section. The outer face of the pipe is in contact with the ambient air at p_a, so the force on the pipe is computed with gauge pressure $p - p_a$.
- Geometry (figure): the liquid enters horizontally along $+\bar{e}_x$ and leaves as a vertical jet along $-\bar{e}_z$ ($\bar{e}_z$ upwards).
- Data: $[\rho,\ c,\ A_i,\ A_o,\ v_o,\ p_i,\ p_a,\ T_i,\ T_o]$. Unknowns: $[v_i,\ F_x,\ F_z,\ \dot{Q}]$.

Symbol	Meaning	SI unit
A_i, A_o	inlet and outlet areas	m^2
v_i, v_o	inlet and outlet velocities	m/s
p_i, p_a	inlet and ambient pressures	Pa
T_i, T_o	inlet and outlet temperatures	K
c	specific heat of the liquid	J/(kg·K)
$\dot{Q}$	heat per unit time into the liquid	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe, $\bar{e}_x$ horizontal (inlet flow direction), $\bar{e}_z$ vertical upwards. Fixed control volume: the liquid inside the pipe, $\bar{v}_c = 0$.
- Surfaces: Σ_i (inlet, $\bar{n} = -\bar{e}_x$, $\bar{v} = v_i \bar{e}_x$), Σ_o (outlet, $\bar{n} = -\bar{e}_z$, $\bar{v} = -v_o \bar{e}_z$), Σ_w (pipe wall, $\bar{v} = 0$).

Force of the liquid and the ambient air on the pipe, and heat through the walls, written on Σ_w:

$$\boxed{\bar{F} = F_x \bar{e}_x + F_z \bar{e}_z = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

$$\boxed{\dot{Q} = -\int_{\Sigma_w} \bar{q} \cdot \bar{n}\, d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_i	$\rho\,v_i\bar{e}_x\cdot(-\bar{e}_x)A_i = -\rho v_i A_i$
Σ_o	$\rho\,(-v_o\bar{e}_z)\cdot(-\bar{e}_z)A_o = +\rho v_o A_o$
Σ_w	0

$$\boxed{v_i = \frac{A_o}{A_i}\,v_o}, \qquad \dot{m} = \rho v_i A_i = \rho v_o A_o \tag{1}$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$\rho v_i\bar{e}_x\,(-v_i)A_i = -\rho v_i^2 A_i\,\bar{e}_x$	$-(p_i-p_a)(-\bar{e}_x)A_i = (p_i-p_a)A_i\,\bar{e}_x$	0
Σ_o	$\rho(-v_o\bar{e}_z)\,(v_o)A_o = -\rho v_o^2 A_o\,\bar{e}_z$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$.

$$-\rho v_i^2 A_i\,\bar{e}_x - \rho v_o^2 A_o\,\bar{e}_z = (p_i - p_a)A_i\,\bar{e}_x - \bar{F} \implies \boxed{F_x = \left[\rho v_i^2 + (p_i - p_a)\right] A_i, \qquad F_z = \rho v_o^2 A_o} \quad (2)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_H\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}\left(cT_i + \frac{v_i^2}{2}\right)$	$+p_i v_i A_i$	0	0 (uniform T)
Σ_o	$+\dot{m}\left(cT_o + \frac{v_o^2}{2}\right)$	$-p_a v_o A_o$	0	0 (uniform T)
Σ_w	0	0 ($\bar{v} = 0$)	0 ($\bar{v} = 0$)	$-\int_{\Sigma_w} \bar{q}\cdot\bar{n}\,d\sigma$

Boxed Σ_w term: $\dot{Q}$. Using (1), $p_i v_i A_i - p_a v_o A_o = (p_i - p_a)\,v_o A_o$.

$$\dot{m}\left[c(T_o - T_i) + \frac{v_o^2 - v_i^2}{2}\right] = (p_i - p_a)\,v_o A_o + \dot{Q} \implies \boxed{\dot{Q} = \rho v_o A_o\left[c(T_o - T_i) + \frac{v_o^2 - v_i^2}{2}\right] + (p_a - p_i)\,v_o A_o} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $F_x > 0$: the pipe is pushed along the incoming flow by the momentum flux and the gauge pressure at the inlet. $F_z > 0$: the jet leaves downwards, so the pipe is pushed upwards.
- (3) splits the heat into the rise of internal energy, the change of kinetic energy and the flow work. If no heat were exchanged and there were no losses, (3) with $\dot{Q} = 0$ and $T_o = T_i$ would reduce to Bernoulli's equation $p_i + \frac{1}{2}\rho v_i^2 = p_a + \frac{1}{2}\rho v_o^2$.

- Dimensions (SI): $[\rho v_i^2 A_i] = [(p_i - p_a) A_i] = \mathrm{N}$; $[\rho v_o A_o \, c \, \Delta T] = (\mathrm{kg/s})(\mathrm{J/(kg\,K)})\,\mathrm{K} = \mathrm{W}$; $[(p_a - p_i) v_o A_o] = \mathrm{Pa \cdot m^3/s} = \mathrm{W}$.
- Numerical illustration (hypothetical data): water, $c = 4186$ J/(kg K), $A_i = 0.01$ m^2, $A_o = 0.004$ m^2, $v_o = 6$ m/s, $p_i = 1.6 \times 10^5$ Pa, $p_a = 101325$ Pa, $T_o - T_i = 0.5$ K give $v_i = 2.4$ m/s, $F_x = 644.35$ N, $F_z = 144$ N and $\dot{Q} = 49187$ W.
- Agreement with the official and class solutions: (1), (2) and (3) are the results of CL6.pdf (which labels the vertical axis y: $F_y = \rho v_o^2 A_o$) and of the class notes.